Dense Unsupervised Learning for Video Segmentation

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NEURAL INFORMATION PROCESSING SYSTEMS

Setup

Goal: Learning dense representations without any supervision for video object segmentation (VOS).

VOS task (test-time):



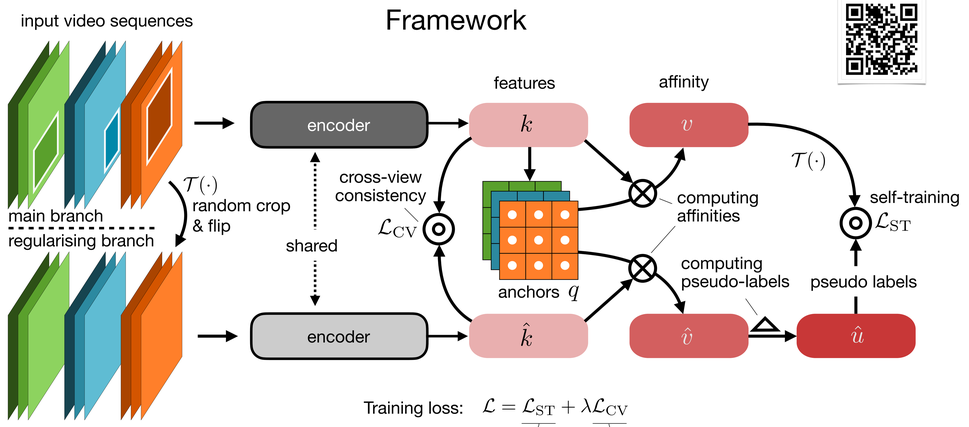
Given object annotation (1 frame)

Propagate in time (video)

Training data: Video sequences without any annotation.

Key assumption: Feature equivariance to similarity transformations (scaling and horizontal flipping).

Framework



Training loss: $\mathcal{L} = \mathcal{L}_{\text{ST}} + \lambda \mathcal{L}_{\text{CV}}$

Self-training

Anchors: $q = \mathcal{G}_N(k)$

Affinities w.r.t. q in both views:

$$\hat{v}_{i,j} = \frac{\exp(\hat{k}_i \cdot q_j / \tau)}{\sum_l \exp(\hat{k}_i \cdot q_l / \tau)}$$

$$v_{i,j} = \frac{\exp(k_i \cdot q_j / \tau)}{\sum_l \exp(k_i \cdot q_l / \tau)}$$

Pseudo-labels: $\hat{u}_i = \underset{j \in \mathcal{N}(i)}{\operatorname{argmax}} \hat{v}_{i,j}$

Self-training loss: $\mathcal{L}_{\text{ST}} = -\sum_{i \notin \mathcal{R}} \mathcal{T}(v_{i,\hat{u}_i})$

Notation

k, $\hat{k}$	dense feature embeddings
$\mathcal{G}_N, \mathcal{G}_M$	grid sampling operator
N, M	grid size
$\mathcal{T}$	random similarity transform
τ	temperature
$\mathcal{R}$	reference frame index set
$\mathcal{N}$	video index set
λ	trade-off scalar

Cross-view consistency

Subsample features:

$$r = \mathcal{G}_M(\mathcal{T}(k))$$

$$\hat{r} = \mathcal{G}_M(\hat{k})$$

Pairwise affinities:

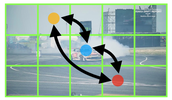
$$h_i = \frac{\exp(r_i \cdot \hat{r}_i / \tau)}{\sum_{l \neq i} \exp(r_i \cdot \hat{r}_l / \tau)}$$

Cross-view loss:

$$\mathcal{L}_{\text{CV}} = -\sum\nolimits_{i \in \mathcal{R}} \log h_i$$

Assumptions

A1 Non-local feature diversity (Jabri et al. [2])



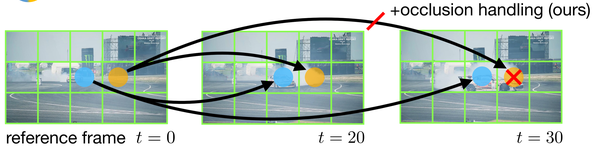
Features are spatially distinguishable

A2 Temporal coherence (Wang et al. [1])



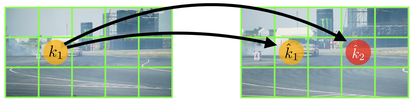
Features from the same video form clusters: $d(k_1, k_2) < d(k_1, k_3)$

A3 Temporal persistence of semantic content (Jabri et al. [2])



Features from the reference frame may be occluded in other frames

A4 Equivariance (ours)



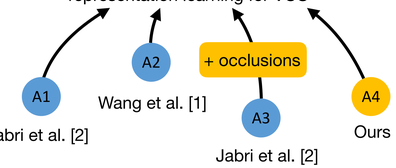
view 1: video n view 2: video n (scaled)

$d(k_1, \hat{k}_1) < d(k_1, \hat{k}_{i \neq 1})$

Features are equivariant to similarity transformations

Contribution

Efficient and effective unsupervised *fully convolutional* representation learning for VOS



Result summary

Accuracy
- state-of-the-art VOS accuracy
- high accuracy even in low-data regime
- consistent across training and testing scenarios

Efficiency
- faster training convergence
- lower computational and memory footprint

Experiments

Metrics

Region similarity (Jaccard): mean $\mathcal{J}_m$ / recall $\mathcal{J}_r$

Contour F-measure: mean $\mathcal{F}_m$ / recall $\mathcal{F}_r$

DAVIS-2017

Method	Train data (hours)	$\mathcal{J}_m\uparrow$	$\mathcal{J}_r\uparrow$	$\mathcal{F}_m\uparrow$	$\mathcal{F}_r\uparrow$	$\mathcal{J}\&\mathcal{F}_m\uparrow$
ContrastCorr [1]	TrackingNet (140)	60.5	—	65.5	—	63.0
Ours		**67.1**	80.9	**71.7**	84.8	**69.4**
CRW [2]	Kinetics (833)	64.8	76.1	70.2	82.1	67.6
Ours		**66.7**	**81.4**	**70.7**	**84.1**	**68.7**
MAST [3]	YouTube-VOS (5)	63.3	73.2	67.6	77.7	65.5
Ours		**67.1**	**81.2**	**71.6**	**84.9**	**69.3**

Qualitative



Efficiency

Method	Train data (hours)	# iterations	duration	GPUs
ContrastCorr [1]	TrackingNet (140)	0.4M	1d	4 x 11Gb
Ours		**0.2M**	1d	**12Gb**
CRW [2]	Kinetics (833)	2M	7d	22Gb
Ours		**0.3M**	**2d**	**12Gb**
MAST [3]	YouTube-VOS (5)	2M	—	22Gb
Ours		**0.1M**	**6h**	**12Gb**

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References

[1] Wang et al., **Contrastive Transformation for Self-supervised Correspondence Learning.** In: AAAI 2021.

[2] Jabri et al., **Space-Time Correspondence as a Contrastive Random Walk.** In: NeurIPS*2020.

[3] Lai et al., **MAST: A Memory-Augmented Self-Supervised Tracker.** In: CVPR 2020.